Module 3

COGNITIVE ASPECTS

What is cognition?

- Thinking, remembering, learning, daydreaming, decision-making, seeing, reading, talking, writing…
- Ways of classifying cognition at a higher level:
  - Experiential vs. reflective cognition (Norman, 1993)
  - Fast vs slow thinking (Kahneman, 2011)

Which involves fast vs slow thinking?

- $2 + 2 =$
- $21 \times 29 =$
- What color eyes do you have?
- How many colors are there in the rainbow?
- How many months in the year have 31 days?
- What is the name of the first school you attended?

Cognitive processes

- Attention
- Perception
- Memory
- Learning
- Reading, speaking and listening
- Problem-solving, planning, reasoning and decision-making

Attention

- Selecting things on which to concentrate at a point in time from the mass of stimuli around us
- Allows us to focus on information that is relevant to what we are doing
- Involves audio and/or visual senses
- Focused and divided attention
  - Enables us to be selective in terms of the mass of competing stimuli, but limits our ability to keep track of all events
- Design recommendation
  - Information at the interface should be structured to capture users' attention, for example, use perceptual boundaries (windows), color, reverse video, sound, and flashing lights

Activity: Find the price for a double room at the Quality Inn in Pennsylvania

Pennsylvania
Bedford Motel/Hotel: Crinaline Courts
(814) 623-9511 S: $118 D: $120
Bedford Motel/Hotel: Holiday Inn
(814) 623-9006 S: $129 D: $136
Bedford Motel/Hotel: Midway
(814) 623-8107 S: $121 D: $126
Bedford Motel/Hotel: Penn Manor
(814) 623-8177 S: $119 D: $125
Bedford Motel/Hotel: Quality Inn
(814) 623-5189 S: $123 D: $128
Bedford Motel/Hotel: Terrace
(814) 623-5111 S: $122 D: $124
Bradley Motel/Hotel: De Soto
(814) 362-3567 S: $120 D: $124
Bradley Motel/Hotel: Holiday House
(814) 362-4511 S: $122 D: $125
Bradley Motel/Hotel: Holiday Inn
(814) 362-4501 S: $132 D: $140
Breezewood Motel/Hotel: Best Western Plaza
(814) 735-4352 S: $120 D: $127
Breezewood Motel/Hotel: Motel 70
(814) 735-4385 S: $116 D: $118

Activity: Find the price of a double room at the Holiday Inn in Columbia

South Carolina

City	Motel/Hotel	Area code	Phone	Rates Single	Rates Double
Charleston	Best Western	803	747-0961	$126	$130
Charleston	Days Inn	803	881-1000	$118	$124
Charleston	Holiday Inn N	803	744-1621	$136	$146
Charleston	Holiday Inn SW	803	556-7100	$133	$147
Charleston	Howard Johnsons	803	524-4148	$131	$136
Charleston	Ramada Inn	803	774-8281	$133	$140
Charleston	Sheraton Inn	803	744-2401	$134	$142
Columbia	Best Western	803	796-9400	$129	$134
Columbia	Carolina Inn	803	799-8200	$142	$148
Columbia	Days Inn	803	736-0000	$123	$127
Columbia	Holiday Inn NW	803	794-9440	$132	$139
Columbia	Howard Johnsons	803	772-7200	$125	$127
Columbia	Quality Inn	803	772-0270	$134	$141
Columbia	Ramada Inn	803	796-2700	$136	$144
Columbia	Vagabond Inn	803	796-6240	$127	$130

Activity

- Tullis (1987) found that the two screens produced quite different results
  - 1st screen: Took an average of 5.5 seconds to search
  - 2nd screen: Took 3.2 seconds to search
- Why, since both displays have the same density of information (31percent)?
- Spacing
  - In the 1st screen, the information is bunched up together, making it hard to search
  - In the 2nd screen, the characters are grouped into vertical categories of information making it easier

Multitasking and attention

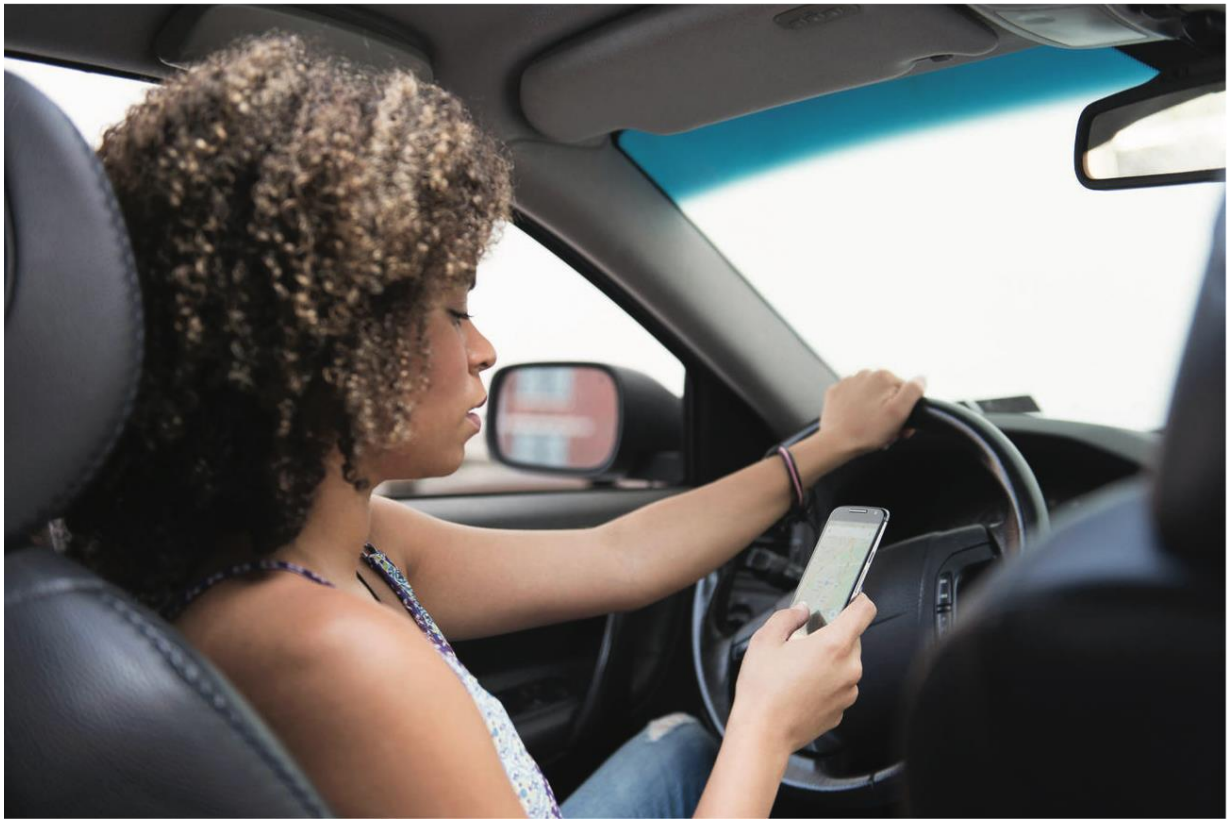
- Is it possible to perform multiple tasks without one or more of them being detrimentally affected?
- Multitasking can cause people to lose their train of thought, make errors, and need to start over
- Ophir et al. (2009) compared heavy vs light multitaskers
  - Heavy multitaskers were more prone to being distracted than those who infrequently multitask
  - Heavy multitaskers are easily distracted and find it difficult to filter irrelevant information

Multitasking at work

It is increasingly common for workers to multitask

- For example, hospital workers have to attend to multiple screens in an operating room that provide new kinds of real-time information
- This requires clinician’s constant attention to check if any data is unusual or anomalous
- Need to develop new attention and scanning strategies

Is it OK to use a phone when driving?



No!

- Driving is very demanding
- Drivers are prone to being distracted
- There is a significant chance of causing accidents
- Drivers' reaction times are longer to external events when talking on the phone in a car (Caird et al., 2018)
- Drivers using their phones rely more on their expectations about what is likely to happen next as conducting a conversation takes up their attention
- Response time is slower to unexpected events (Briggs et al., 2018)
- Drivers often try to imagine what the other person's face is like−the person to whom they are speaking
  - Doing so competes with the processing resources needed to enable them to notice and react to what is in front of them

Are hands-free phones safer to use when driving?

- No, as same type of cognitive processing is happening when talking
- The same thing happens when talking with front seat passenger
  - But both can stop in mid-sentence if a hazard is spotted allowing the driver to switch immediately to the road
  - So, it’s less dangerous talking to a front seat passenger than a remote person
  - A remote person on the end of a phone is not privy to what the driver is seeing and will carry on the conversation when there is a hazard
  - This makes it difficult for the driver to switch all their attention to the road

Design implications for attention

- Context: Make information salient when it needs to be attended to at a given stage of a task
- Use techniques to achieve this:
  - For example, color, ordering, spacing, underlining, sequencing, and animation
- Avoid cluttering visual interfaces with too much information
- Consider designing different ways to support effective switching and returning to an interface

Perception

- How information is acquired from the world and transformed into experiences
- Obvious implication is to design representations that are readily perceivable, for instance:
  - Text should be legible
  - Icons should be easy to distinguish and read

Is color contrast good? Find Italian

Black Hills Forest	Peters Landing	Jefferson Farms	Devlin Hall
Cheyenne River	Public Health	Psychophysics	Positions
Social Science	San Bernardino	Political Science	Hubard Hall
South San Jose	Moreno Valley	Game Schedule	Fernadino Beach
Badlands Park	Altamonte Springs	South Addision	Council Bluffs
Juvenile Justice	Peach Tree City	Cherry Hills Village	Classical Lit
Results and Stats	Highland Park	Creative Writing	Sociology
Thousand Oaks	Manchesney Park	Lake Havasu City	Greek
Promotions	Vallecito Mts.	Engineering Bldg	Wallace Hall
North Palermo	Rock Falls	Sports Studies	Concert Tickets
Credit Union	Freeport	Lakewood Village	Public Radio FM
Wilner Hall	Slaughter Beach	Rock Island	Children's Museum
Performing Arts	Rocky Mountains	Deerfield Beach	Writing Center
Italian	Latin	Arlington Hill	Theater Auditions
Coaches	Pleasant Hills	Preview Game	Delaware City
McKees Rocks	Observatory	Richland Hills	Scholarships
Glenwood Springs	Public Affairs	Experts Guide	Hendricksville
Urban Affairs	Heskett Center	Neff Hall	Knights Landing
McLeansboro	Brunswick	Grand Wash Cliffs	Modern Literature
Experimental Links	East Millinocket	Indian Well Valley	Studio Arts
Graduation	Women's Studies	Online Courses	Hughes Complex
Emory Lindquist	Vacant	Lindquist Hall	Cumberland Flats
Clinton Hall	News Theatre	Fisk Hall	Central Village
San Luis Obispo	Candlewood Isle	Los Padres Forest	Hoffman Estates

Are borders and white space better?
Find French

Webmaster	Curriculum	Student Life	Dance
Russian	Emergency (EMS)	Accountancy	Gerontology
Athletics	Statistics	McKnight Center	Marketing
Go Shockers	Award Documents	Council of Women	College Bylaws
Degree Options	Language Center	Commute	Why Wichita?
Newsletter	Future Shockers	Small Business	Tickets
Geology	Intercollegiate	Thinker & Movers	Career Services
Manufacturing	Bowling	Alumni	Doers & Shockers
Management	Wichita Gateway	Foundations	Core Values
UCATS	Transfer Day	Corbin Center	Grace Wilkie Hall
Alumni News	Job Openings	Jardine Hall	Strategic Plan
Saso	Live Radio	Hugo Wall School	Medical Tech
Educational Map	Beta Alpha Psi	Staff	Softball, Men's
Physical Plant	Liberal Arts	Aerospace	McKinley Hall
Graphic Design	Counseling	Choral Dept.	Email
Non Credit Class	Biological Science	Alberg Hall	Dental Hygiene
Media Relations	Duerksen Fine Art	French	Tenure
Advertising	EMT Program	Spanish	Personnel Policies
English	Religion	Parents	Instrumental
Graduate Complex	Art Composition	Wrestling	Nursing
Music Education	Physics	Philosophy	Opera
Advising Center	Entrepreneurship	Wichita Lyceum	Sports History
Medical School	Koch Arena	Fairmount Center	Athletic Dept.
Levitt Arena	Roster	Women's Museum	Health Plan

Activity: Which is the easiest to read and why?

What is the time?

What is the time?

What is the time?

What is the time?

What is the time?

Design implications

- Icons should enable users to *distinguish* their meaning readily
- Bordering and spacing are effective visual ways of grouping information
- Sounds should be audible and distinguishable
- Research proper color contrast techniques when designing an interface:
  - Yellow on black or blue is fine
  - Yellow on green or white is a no-no
- Haptic feedback should be used judiciously

Memory

- Involves recalling various kinds of knowledge that allow people to act appropriately
  - For example, recognizing someone's face or remembering someone's name
- First encode and then retrieve knowledge
- We don't remember everything−it involves filtering and processing what is attended to
- Context is important as to how we remember (that is, where, when, how, and so on)
- We recognize things much better than being able to recall things
- We remember less about objects that we have photographed than when we observe them with the naked eye (Henkel, 2014)

Processing in memory

- Encoding is first stage of memory
  - Determines which information is attended to in the environment and how it is interpreted
- The more attention paid to something…
- The more it is processed in terms of thinking about it and comparing it with other knowledge…
- The more likely it is to be remembered
  - For example, when learning about HCI, it is much better to reflect upon it, carry out exercises, have discussions with others about it, and write notes than just passively read a book, listen to a lecture or watch a video about it

Context is important

- Context affects the extent to which information can be subsequently retrieved
- Sometimes it can be difficult for people to recall information that was encoded in a different context:
  - “You are on a train and someone comes up to you and says hello. You don’t recognize him for a few moments, but then realize it is one of your neighbors. You are only used to seeing your neighbor in the hallway of your apartment building, and seeing him out of context makes him difficult to recognize initially”

The problem with the classic ‘7,+ or − 2’

- George Miller’s (1956) theory of how much information people can remember
- People’s immediate memory capacity is very limited to 7, + or − 2
- Has been applied in interaction design when considering how many options to display
- But is it a good use of a theory in HCI?
- Is it helpful?

When creating an interface, should the designer…

- Present only 7 options on a menu
- Display only 7 icons on a tool bar
- Have no more than 7 bullets in a list
- Place only 7 items on a pull down menu
- Place only 7 tabs on the top of a website page?
- Not necessarily…

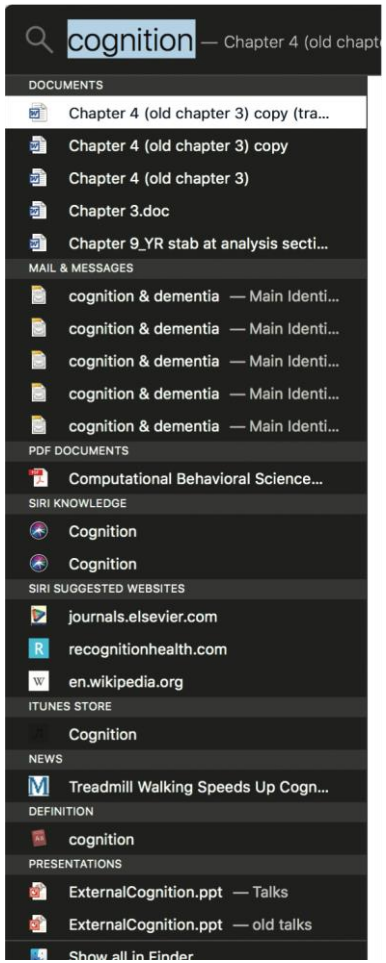
The reason is…

- People can scan lists of bullets, tabs, and menu items for the one they want
- They don’t have to recall them from memory, having only briefly heard or seen them
- So you can have more than nine at the interface
  - For instance, history lists of websites visited
- Sometimes a small number of items is good
  - For example, smart watch displays
- Depends on task and available screen estate

Personal Information management

- Bergman and Whittaker, three interdependent processes model (2016) to help people manage their stuff:
  I. How to decide what stuff to keep
  II. How to organize it when storing
  III. Which strategies to use to retrieve it later
- Most common approach is to use folders and naming
- Strong preference for scanning across and within folders when looking for something
- Search engines only helpful if you know the name of the file
- Smart search engines help with listing relevant files for partial name or when type in first letter

Apple’s Spotlight search tool



Memory load

- Online/mobile and phone banking now require users to provide multiple pieces of information to access their account
  - For instance, ZIP code, birthplace, a memorable date, first school attended
  - Known as multifactor authentication (MFA)
- Why?
  - Increased security concerns
- Password managers, such as LastPass, have been developed that require only one master password
  - Reduces stress and memory load on users
- Passwords could become extinct with the widespread use of biometrics and computer vision algorithms

Digital Forgetting

- When might you wish to forget something that is online?
  - When you break up with a partner
  - Emotionally painful to be reminded of them through shared photos, social media, and so on.
- Sas and Whittaker (2013) suggest ways of harvesting and deleting digital content
  - For example, making photos of ex into an abstract collage
  - Helps with closure

Memory aids

- SenseCam, developed by Microsoft Research Labs (now Autographer)
  - A wearable device that intermittently takes photos without any user intervention while worn
  - Digital images taken are stored and revisited using special software
  - Has been found to improve people’s memory, especially those suffering from dementia
- Other aids include RemArc, which triggers long-term memory using old BBC materials

SenseCam

Design implications

- Reduce cognitive load by avoiding long and complicated procedures for carrying out tasks
- Design interfaces that promote recognition rather than recall
- Provide users with various ways of labelling digital information to help them easily identify it again
  - For example, folders, categories, color, flagging, and time stamping

Learning

- Involves the accumulation of skills and knowledge involving memory
- Two main types:
  - Incidental learning (for example, recognizing people's faces, what you did today)
  - Intentional learning (for instance, studying for an exam, learning to cook)
  - Intentional learning is much harder!
  - Many technologies have been developed to help (for example, multimedia, animations, VR)
- People find it hard to learn by following instructions in a manual
- People prefer to learn by doing

Design implications

- Design interfaces that encourage exploration
- Design interfaces that constrain and guide learners
- Dynamically linking concepts and representations can facilitate the learning of complex material

Reading, speaking, and listening

The ease with which people can read, listen, or speak differs:

- Many prefer listening to reading
- Reading can be quicker than speaking or listening
- Listening requires less cognitive effort than reading or speaking
- Dyslexics have difficulties understanding and recognizing written words

Applications

- Voice user interfaces allow users to interact with them by asking questions
  - For example, Google Voice, Siri, and Alexa
- Speech-output systems use artificially-generated speech
  - For instance, written text-to-speech systems for the visually impaired
- Natural-language systems enable users to type in questions and give text-based responses
  - Such as, chatbots

Design implications

- Speech-based menus and instructions should be short
- Accentuate the intonation of artificially generated speech voices
  - They are harder to understand than human voices
- Provide opportunities for making text large on a screen

Problem-solving, planning, reasoning, and decision-making

- All these processes involve *reflective* cognition
  - For example, thinking about what to do, what the options are, and the consequences
- Often involves conscious processes, discussion with others (or oneself), and the use of artifacts
  - Such as maps, books, pen and paper
- May involve working through different scenarios and deciding which is best option
- Weighing up alternatives

Design implications

- Provide information and help pages that are easy to access for people who wish to understand more about how to carry out an activity more effectively (for example, web searching)
- Use simple and memorable functions to support rapid decision-making and planning

Dilemma

- The app mentality is making it worse for people to make their own decisions because they are becoming risk averse (Gardner and Davis, 2013)
  - Instead, they now rely on a multitude of apps
  - This makes them increasingly anxious
  - They are unable to make decisions by themselves
  - They need to resort to looking up info, getting other's opinions on social media, and comparing notes
- Do you agree?
- Did it happen to you when deciding which university/school to attend?

Cognitive frameworks

- These are used to explain and predict user behavior at the interface
  - Based on theories of behavior
  - Focus is on mental processes that take place
  - Also use of artifacts and representations
- Most well known are:
  - Mental models
  - Gulfs of execution and evaluation
  - Distributed cognition
  - External and embodied cognition

Human Reaction and Prediction of Cognitive Performance

- We can also, to some degree, predict how humans will react and perform in response to a particular human interface design.
- We can consider two aspects of human performance: one that is cognitive and the other ergonomic.
- “Gulf of Execution/Evaluation”
  - the gap between the expected and actual
- The mismatch between the user’s mental model and the task model employed by the interactive system creates the “gulf.”

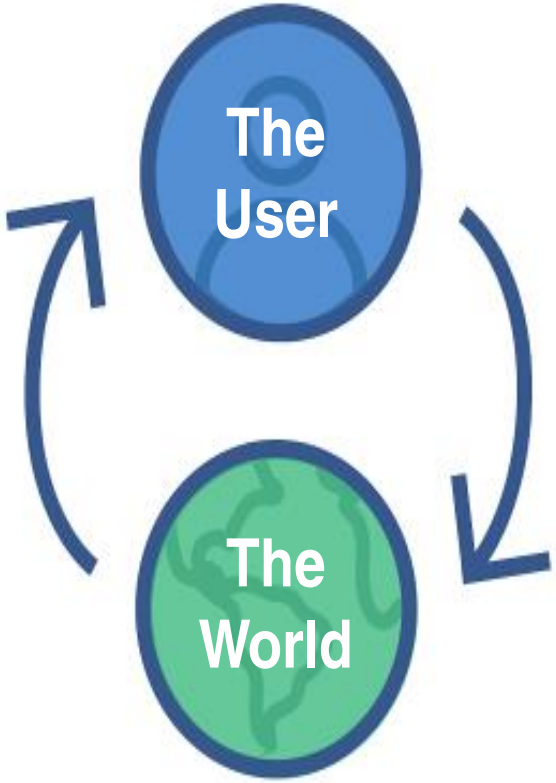
Gulfs of execution and evaluation

- The ‘gulfs’ explicate the gaps that exist between the user and the interface
- The gulf of execution
  - The distance from the user to the physical system
- The gulf of evaluation
  - The distance from the physical system to the user
- Bridging the gulfs can reduce cognitive effort required to perform tasks
- Can reveal whether interface increases or decreases cognitive load and whether it is obvious what to do next (Norman, 1986; Hutchins et al, 1986)

Bridging the gulfs

Gulf of Evaluation

What's the current system state?



Gulf of Execution

How do I use this system?

- When the task model and interface structure of the interactive system maps well to the expected mental model of the user, the task performance will be very fluid.

- Memory capacity also influences the interactive performance greatly.

- There are largely two types of memory in the human cognitive system:
    1. The short term(working memory) and
    2. The long term.

- The short-term memory is also sometimes known as the working memory, in the sense that it contains (changing) memory elements meaningful for the task at hand (or chunks).

Predictive Performance Assessment: GOMS

- Many important cognitive activities have been analyzed in terms of their typical approximate process time.
- With a Task-Sequence Model, one might be able to quantitatively estimate the time taken to complete a given task and, therefore, make an evaluation with regard to the original performance requirements.
- Framework called GOMS (Goals, Operators, Methods, and Selection).

Table 3.1 Estimates of Time Taken for Typical Desktop Computer Operations from GOMS

TYPE OF OPERATION	TIME ESTIMATE
K: Keyboard input	Expert: 0.12 s Average: 0.20 s Novice: 1.2 s
T(*n*): Type *n* characters	$280 \times n$ ms
P: Point with mouse to something on the display	1100 ms
B: Press or release mouse button	100 ms
BB: Click a mouse button (press and release)	200 ms
H: Home hands, either to the keyboard or mouse	400 ms
M: Thinking what to do (mental operator)	1200 ms (can change)
W(*t*): Waiting for the system (to respond)	*t* ms

Table 3.2 Estimates of Time Taken for Two Task Models of "Deleting a File"

DELETING A FILE			
DESIGN 1		DESIGN 2[a]	
1. Point to file icon	P	1. Point to file icon	P
2. Click mouse button	BB	2. Click mouse button	BB
3. Point to file menu	P	3. Move hand to keyboard	M
4. Press and hold mouse button	B	4. Hit command key: command-T	KK
5. Point to DELETE item	P	5. Move hand back to mouse	H
6. Release mouse button	B		
7. Point to original window	P		
Total time = 4.8 s		Total time = 2.66 s	

Sensation and Perception of Information

Mental Models

Mental models

- Users develop an understanding of a system through learning about and using it
- Knowledge is sometimes described as a mental model:
  - How to use the system (what to do next)
  - What to do with unfamiliar systems or unexpected situations (how the system works)
- People make inferences using mental models of how to carry out tasks

What are Mental Models?

A persons thought process of understanding the world.

It allows people to make assumption about how things work and unconsciously influence our behavior and decision making.

How are mental models constructed?

- Affordances
- Constraints
- Mappings
- Positive transfer
- Cultural associations/standards
- Instructions
- Interactions

Good Example: Wheel bag

Wheel Bag

- Affordances:

Handle to carry or lift the bag.

Big compartments for clothes and small for accessories.

- Constraints:

Moving zip in one direction to open and close the bag.

- Mappings:

Placement of the side handle to use the wheel for moving.

- Positive transfer:

Learnt easily when in contact

Bad example: Watch

Bad example: Watch

- Affordances:
- Four push buttons, not clear what they do.
- Constraints and mapping:
- No visible relation between buttons and the end-result of their actions.
- Negative transfer:
- Little association with analog watches.

More mental models

- Craik (1943) described mental models as:
  - Internal constructions of some aspect of the external world enabling predictions to be made
- Involves unconscious and conscious processes
  - Imagery and analogies are activated
- Deep versus shallow models
  - For example, how to drive a car and how it works

Everyday reasoning and mental models

(a) You arrive home on a cold winter's night to a cold house. How do you get the house to warm up as quickly as possible? Set the thermostat to be at its highest or to the desired temperature?

(b) You arrive home starving hungry. You look in the fridge and find all that is left is an uncooked pizza. You have an electric oven. Do you warm it up to 375 degrees first and then put it in (as specified by the instructions) or turn the oven up higher to try to warm it up quicker?

Heating up a room or oven that is thermostat-controlled

- Many people when asked (a) choose the first option
- Why?
  - They think it will heat the room up quicker
  - General valve theory, where ‘more is more’ principle is generalized to different settings (for instance, gas pedal, gas cooker, tap, radio volume)
  - But it is a wrong mental model for thermostats based on on-off switch model
- Many people when asked (b) choose the first option
  - Electric ovens work on the same principle as thermostats
- Most of us have erroneous mental models (Kempton, 1996)

Erroneous mental models

- Lots of people hit the button for elevators and pedestrian crossings at least twice
  - Why? Think it will make the lights change faster or ensure that the elevator arrives!
- What kinds of mental models do users have for understanding how interactive devices work?
  - Poor, often incomplete, easily confusable, based on inappropriate analogies and superstition (Norman, 1983)

- First we look at the input side (i.e., the human sensory system). Humans are known to have at least five senses.

- Among them, those that would be relevant to HCI (at least for now) are the modalities of visual, aural, haptic (force feedback), and tactile sensation.

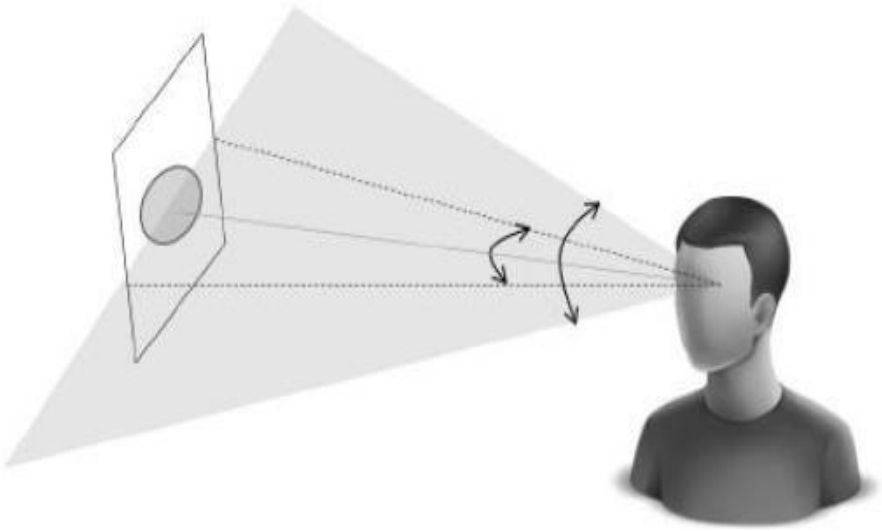
1. Visual

- Over 40% of the human brain is said to be involved with the processing of visual information.

- As already mentioned, the parameters of the visual interface design and display system will have to conform to the capacity and characteristics of the human visual system.

Visual and Display Parameters

- *Field of view (FOV) - Angle*
- *Viewing distance*
- *Display field of view*
- *Pixel*
- *Display resolution*
- *Visual acuity – (Power of Sight*



Color, Brightness, and Contrast

- Attributes of visual quality are
  - **Brightness**- amount of light energy emitted by the object
  - **Color**-Human response to different wavelengths of light
  - **Contrast**- Relative difference in brightness or color between two visual objects

II. Aural (Sense of hearing)

- The aural modality (sound) is perhaps the most prevalent mode for information feedback.
- The actual form of sound feedback can be roughly divided into three types:

- (a) Simple beep like sounds,
- (b) Short symbolic sound bytes known as *earcons* (e.g., the paper-crunching sound when a file is inserted into the trashcane for deletion), and
- (c) Relatively longer “as is” sound feedback that is replayed from recordings or synthesis

Aural Display Parameters

- ***Intensity*** (amplitude)- volume
- ***Phase*** - refers to the time differences among sound waves

Table 3.3 Examples of Different Sounds and Their Typical Intensity Levels in Decibels

INTENSITY (DB)	DESCRIPTION
0	Weakest sound audible
30	Whisper
50	Office environment
60	Normal conversation
110	Rock band
130	Pain threshold

Aural Modality as Input Method

- Keyword Recognition &
- Natural Language Understanding

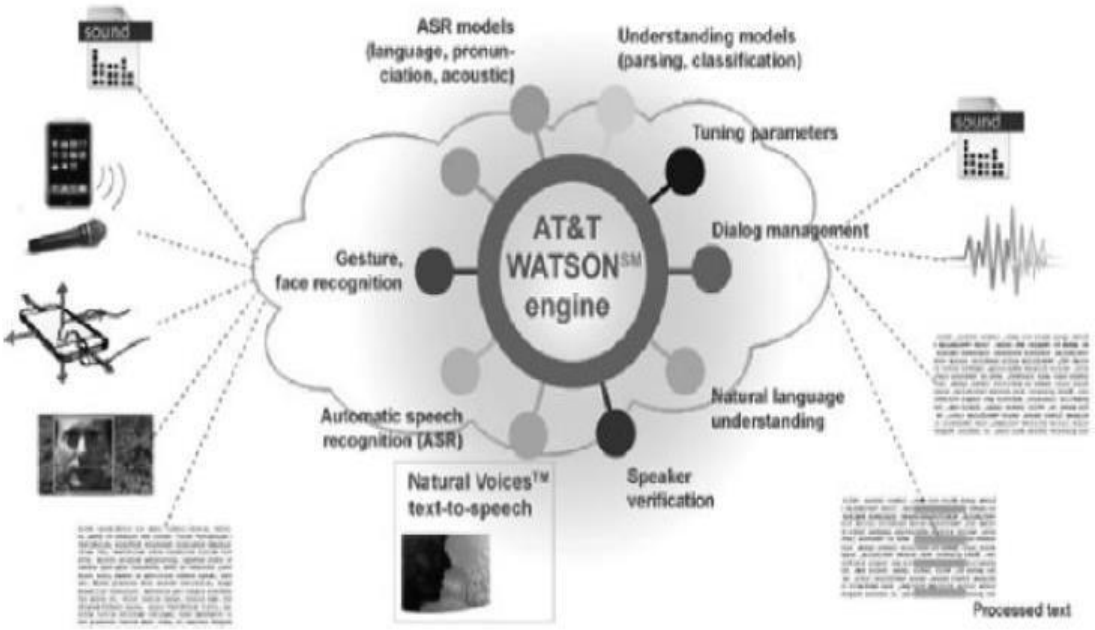


Figure 3.14 A high-quality natural-language-based interaction through the cloud. The smart-media client devices would send the captured sentence (in voice or text), and a correct and intelligent response is given back in real time. (From http://www.research.att.com/project/WATSON.)

III. Tactile and Haptic

- ***haptic*** is defined to be the modality that takes advantage of touch by applying forces, vibrations, or motions to the user.
- ***tactile*** for sensing different types of touch (e.g., texture, light pressure/contact, pain, vibration, and even temperature) through our skin.

Tactile Display Parameters

- *Tactile resolution*: The Skin Sensitivity To Physical Objects
- *Vibration frequency*: Rapid movement
- *Pressure threshold*: The Lightest Amount Of Pressure Humans Can Sense Is Said To Be About 1000 N/M2.

Haptic and Haptic Display Parameters

- Haptic feedback adds a more apparent physical dimension to interaction.
- Force feedback and movement is felt
- In general, due to their mechanical nature, “robotic” haptic devices are not yet very practical.
- They tend to be heavy, clunky, dangerous, and take up a large volume.

Two types of haptic systems:

(a) Grounded and (b) Body worn

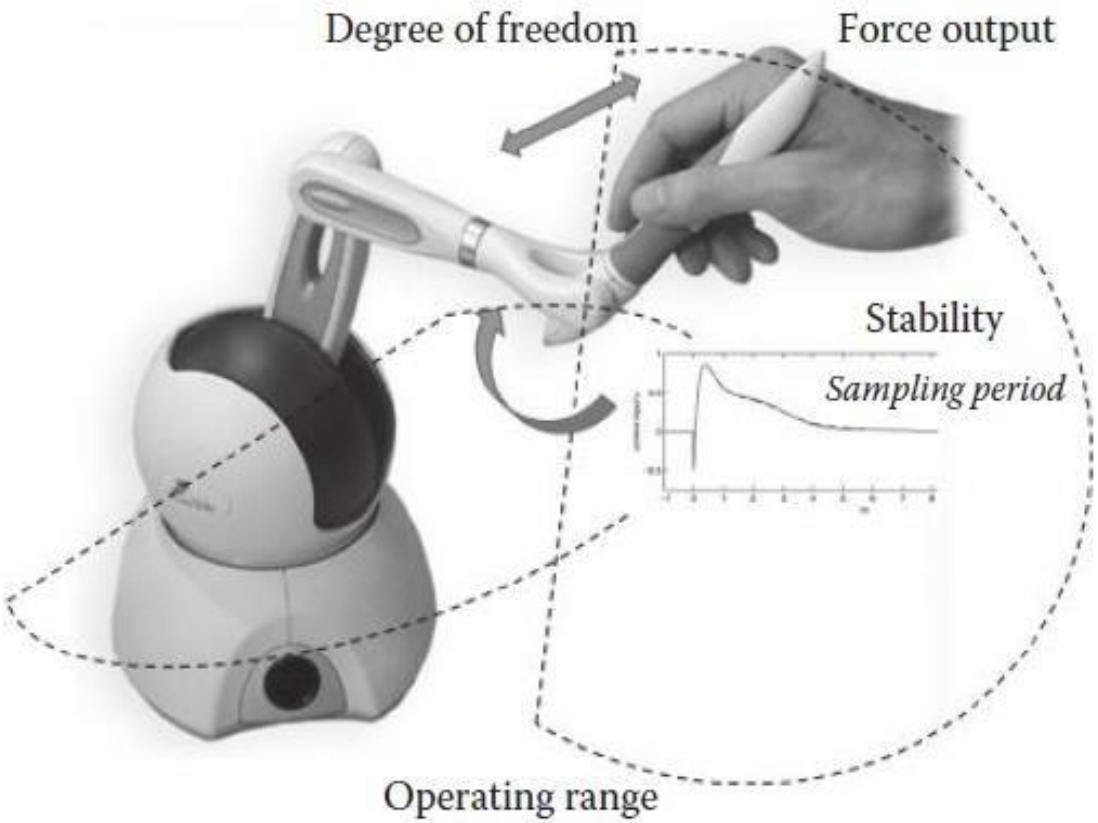


Figure 3.18 Important parameters for a haptic display system.

Multimodal Interaction

- By employing more than one modality, interfaces can become more effective
- Multimodal interfaces are gaining popularity with the ubiquity of multimedia devices.
- *Complementary:*
  - *Different modalities can assume different* roles and act in a complementary fashion to achieve specific interaction objectives.
  - For example, an motor organs can signify the **arrival of a phone call** while the **visual displays the caller's name.**

- *Redundant:* ***Different modality input methods*** *or feedback can be used to* ***ensure a reliable achievement*** *of the interaction objective.*

  For instance, the ***ring of a phone call*** *can be* ***simultaneously*** *aural and tactile to strengthen the pick-up probability.*

- *Alternative:*

  Providing users with ***alternative ways to interact*** *gives people more choices.*

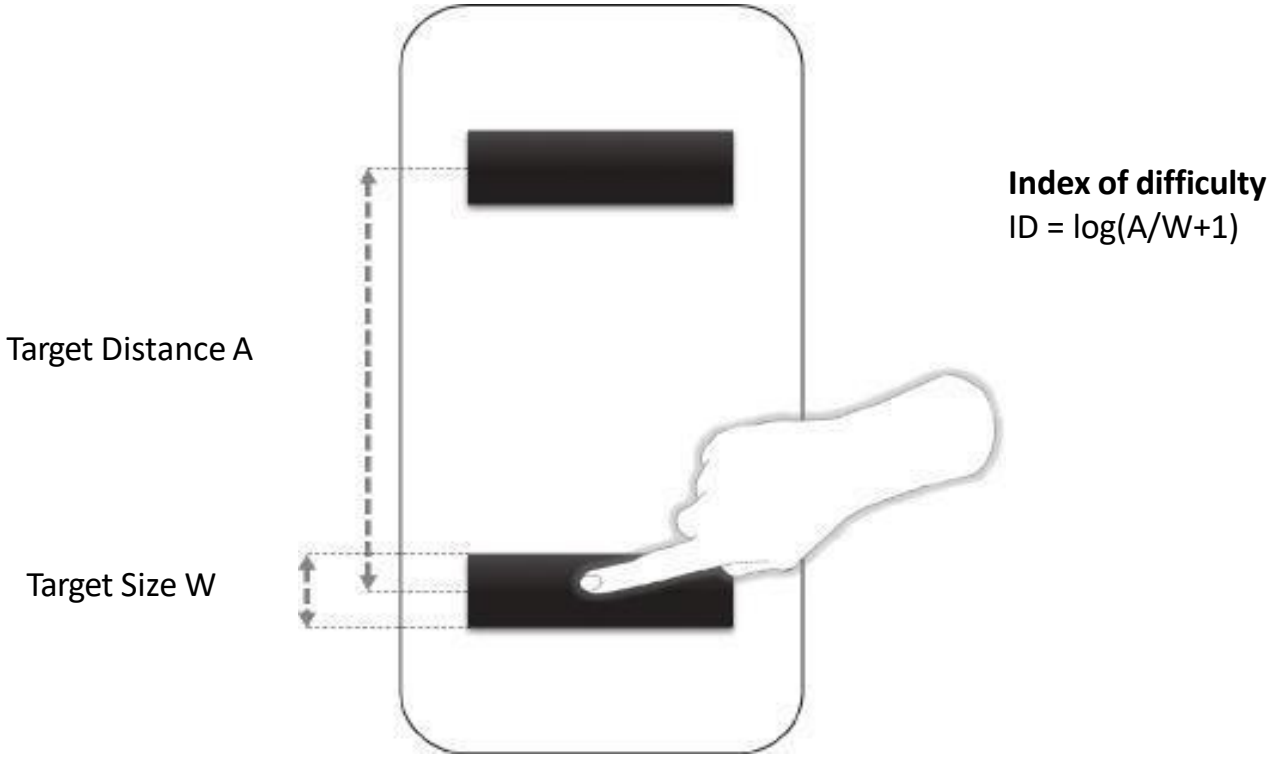
  For instance, a ***phone call can be made*** *either* ***by touching*** *a button or* ***by speaking the callee's name****, thereby promoting convenience and usability.*

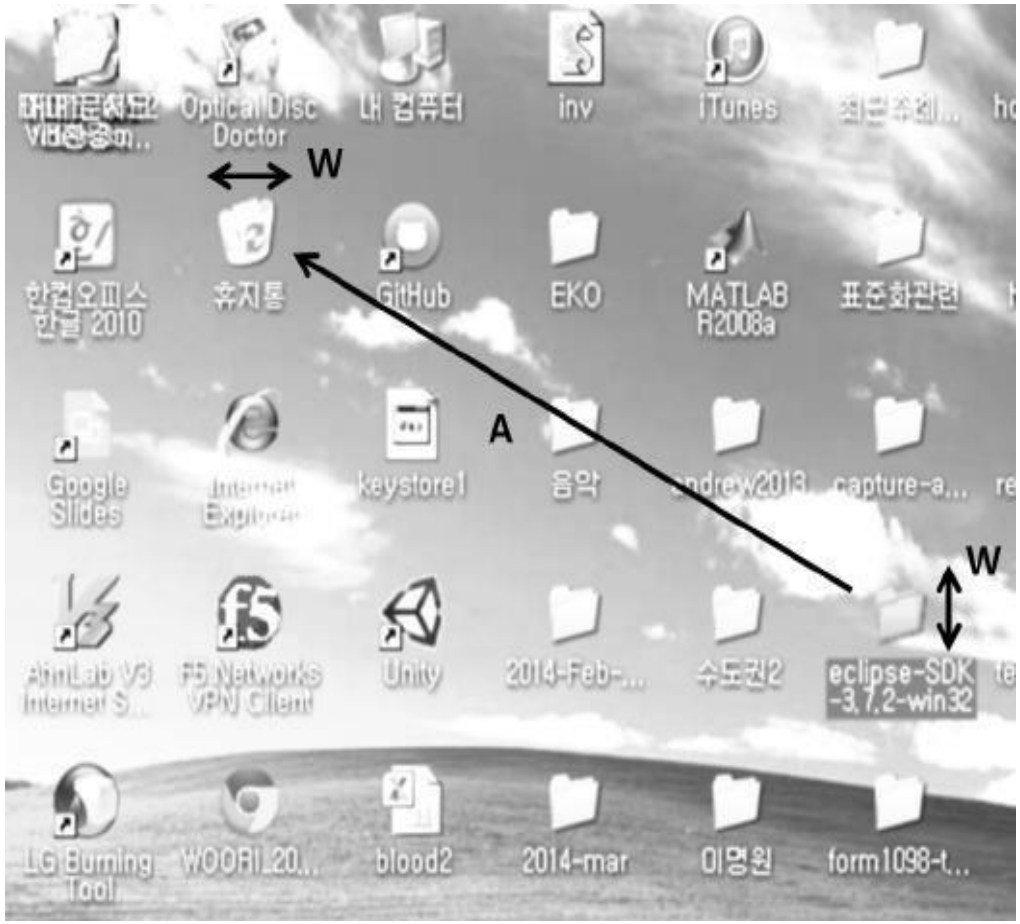
Human Body Ergonomics

- Design of interfaces(interaction devices) for
- comfort and High performance according to the **physical mechanisms of the human body**.
- **Fitts's Law:**

Model of human movement that predicts the **time required** to rapidly move to a target area as a function of the distance to and the size of the target.

- MT = a + b * ID and ID = log(A/W + 1)


Target Distance A
Target Size W
Index of difficulty
ID = log(A/W+1)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eclipse-SDK -3.7.2-win32
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- Motor Control:

***Isometric* devices**, where the movement of the device directly translates to the **movement** in the display (or virtual space).

Nonisometric devices, are those that control the movement in the display in principle with something else such as force, thus possibly with **no movement** input at all.

Others

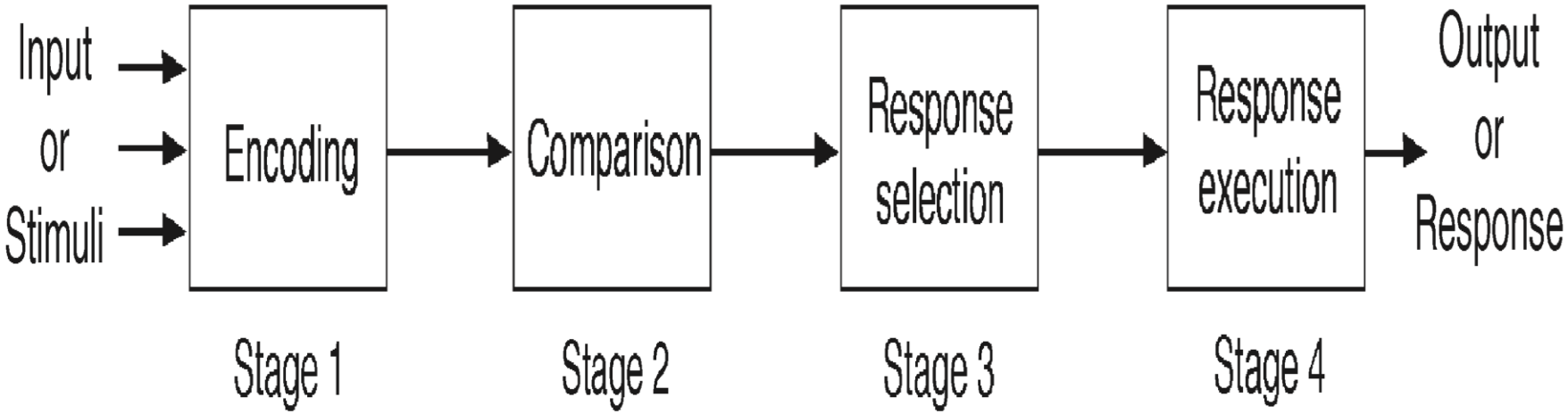
- There are many cognitive, perceptual, and ergonomic issues that have been left out.
- Learning and adaptation
- Modalities other than the “big three” (visual/aural/haptictactile), such as gestures, facial expression, brain waves, physiological signals (electromyogram, heart rate, skin conductance), gaze, etc.
- Aesthetics and emotion
- Multitasking

How can UX be designed to help people build better mental models?

- Clear and easy to use instructions
- Appropriate tutorials and contextual sensitive guidance
- Provide online videos and chatbot windows when needing help
- Transparency: to make interfaces intuitive to use
- Affordances of what actions an interface allows
  - For example, swiping, clicking, or selecting

Information processing

- Conceptualizes human performance in metaphorical terms of information processing stages



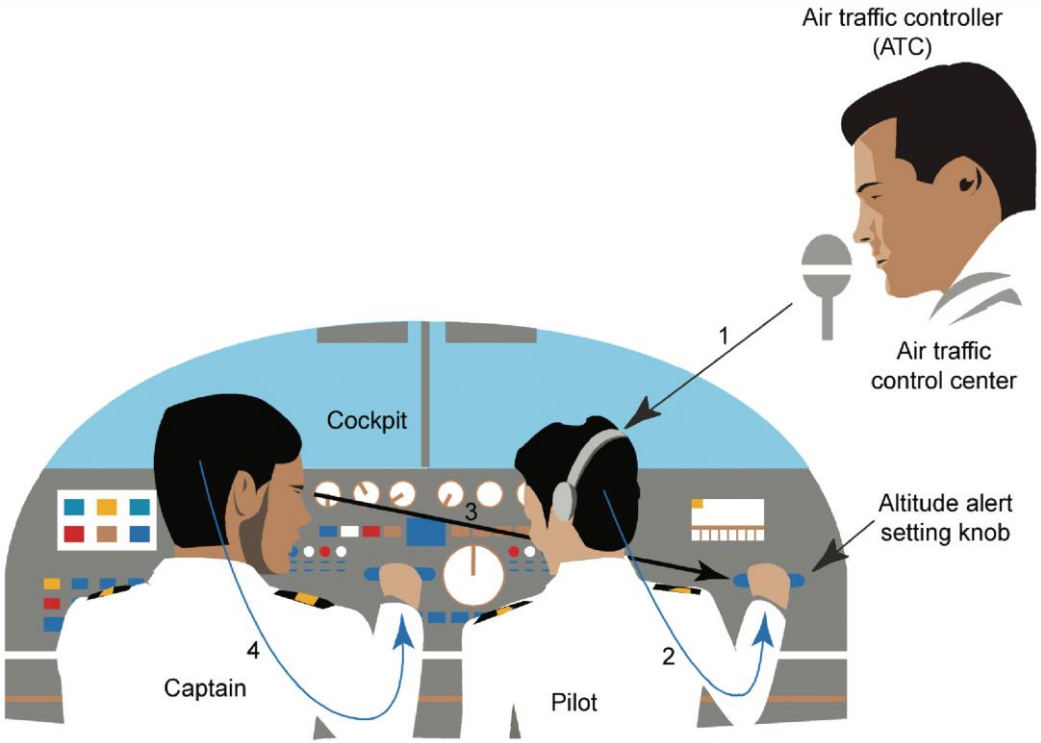
Limitations

- Based on modeling mental activities that happen exclusively inside the head
- Do not adequately account for how people interact with computers and other devices in real world

Distributed cognition

- Concerned with the nature of cognitive phenomena across individuals, artifacts, and internal and external representations (Hutchins, 1995)
- Describes these in terms of propagation across representational state
- Information is transformed through different media (computers, displays, paper, heads)

A cognitive system for ATC



Propagation of representational states:

1 ATC gives clearance to pilot to fly to higher altitude (verbal)

2 Pilot changes altitude meter (mental and physical)

3 Captain observes pilot (visual)

4 Captain flies to higher altitude (mental and physical)

What’s involved

- The distributed problem-solving that takes place
- The role of verbal and non-verbal behavior
- The various coordinating mechanisms that are used (for example, rules and procedures)
- The communication that takes place as the collaborative activity progresses
- How knowledge is shared and accessed

External cognition

- Concerned with explaining how we interact with external representations (such as maps, notes, and diagrams)
- What are the cognitive benefits and what processes involved
- How they extend cognition
- What technologies can we develop to help people carry out complex tasks (for example, learning, problem solving, and decision-making)?

Externalizing to reduce memory load

- Examples include the use of diaries, reminders, calendars, notes, shopping lists, to-do lists
  - Written to remind us of what to do
- Post-its, piles, marked emails are used to:
  - Where placed indicates priority of what to do
- External representations:
  - Remind us that we need to do something (for example, to buy something for mother's day)
  - Remind us of what to do (for instance, buy a card)
  - Remind us when to do something (for example, send a card by a certain date)

Annotation and cognitive tracing

- Annotation involves modifying existing representations through making marks
  - For example, crossing off, ticking, and underlining
- Cognitive tracing involves externally manipulating items into different orders or structures
  - For instance, playing Scrabble or cards

Design implication

- Provide external representations at the interface that can reduce memory load and facilitate computational offloading
  - For example, information visualizations have been designed to allow people to make sense and rapid decisions about masses of data